

Introduction

Relations are a critical part of mathematics: They are how we compare two potentially different objects. Inequalities, functions, orders, and equivalence all arise from relations!

Learning Objectives

- Explain how contradiction proofs work and prove basic results using contradiction.
- Check whether a given relation is reflexive, symmetric, etc.
- Define an equivalence relation and equivalence classes.
- Discuss the special properties of equivalence relations, such as motivating its definition, or describing how equivalence relations partition a set.
- Define a weak (strong) order relation, define maximal and minimal elements of a partially ordered set, and define the (least) upper bound and the (greatest) lower bound for a set.
- Prove basic facts about equivalence relations and partially ordered sets.

Contradiction

Let P and Q be propositions and consider the problem of showing that the statement $T = P \Rightarrow Q$ is true. A proof by contradiction proceeds by assuming that T is false (or $\neg T$ is true), and showing that something bad happens. More specifically, if R is some other proposition which may not be directly related to P or Q , then our goal is to show that

$$\neg T \Rightarrow (R \wedge \neg R) \tag{1}$$

is a true statement. Here we recall that for any predicate R , $R \wedge \neg R$ is always false, so we have shown that the assumption that $\neg T$ is true leads to a contradiction.

We can use a truth table to verify that the truth of T perfectly corresponds with the truth of (1). Indeed

T	R	$\neg T$	$R \wedge \neg R$	$\neg T \Rightarrow (R \wedge \neg R)$
T	T	F	F	T
T	F	F	F	T
F	T	T	F	F
F	F	T	F	F

Hence T and $\neg T \Rightarrow (R \wedge \neg R)$ are logically equivalent, and one can prove that T is true by showing that Equation (1) is true.

Proposition 1

In the decimal expansion of π , at least one of the digits $\{0, 1, 2, \dots, 8, 9\}$ occurs infinitely often.

Proof. For the sake of contradiction, assume that each of the above digits occurs only finitely many times in the decimal expansion of π . Let N_i be the number of times the digit i appears, so that the decimal expansion of π consists of

$$N_0 + N_1 + \dots + N_9$$

digits. As each N_i is finite, so too is this sum. If π has only a finite decimal expansion, it is necessarily rational. Since we know that π is not rational, this is a contradiction and we conclude that some digit must occur infinitely often. $\square$

Note that this proof is not constructive: we do not know which digit occurs infinitely often. In fact, it is an open problem whether each digit occurs infinitely often.

Exercise 2 Show that there is no smallest positive real number.

Exercise 3 If A and B are sets, show that $A \cap (B \setminus A) = \emptyset$.

Proposition 4

The number $\sqrt{2}$ is irrational.

Proof. For the sake of contradiction, assume that $\sqrt{2}$ is rational and write $\sqrt{2} = p/q$ where $\gcd(p, q) = 1$; that is, p/q is in lowest terms. Hence $q\sqrt{2} = p$ and by squaring both sides we get

$$2q^2 = p^2.$$

Notice that $2q^2$ is even, and so therefore p^2 must also be even. By a previous result, we know that p is therefore also even, so $p = 2k$ for some $k \in \mathbb{Z}$. Substituting this back into our equation, we get

$$2q^2 = (2k)^2 = 4k^2 \quad \Leftrightarrow \quad q^2 = 2k^2$$

so that similarly, q^2 is even. This implies that q is even, so $q = 2\ell$. However, this is a contradiction. We assumed that p and q were written in lowest terms, but have demonstrated that both are even. Hence $\sqrt{2}$ is not rational and so must be irrational. $\square$

Here is a slightly more involved proof by contradiction.

Example 5

Show that there are no natural solutions to the equation $x^2 - 4y^2 = 7$.

Solution. Suppose for the sake of contradiction that a solution exists. Note that we can factor the left hand side, giving $x^2 - 4y^2 = (x - 2y)(x + 2y) = 7$. Since 7 is prime, its two factors are either $-1, -7$ or $1, 7$, but we can throw away the negative factors since $x + 2y > 0$.

Thus we either have $x - 2y = 1$ and $x + 2y = 7$, or $x - 2y = 7$ and $x + 2y = 1$. In either case, if we add the two equations together we get $x = 4$. In the first case, this implies that $y = 3/2$ which is not possible. In the latter case, $y = -3/2$, which is also not possible. Hence we've arrived at a contradiction, and no solutions can exist. ■

Exercise 6 Show that $x^3 + x^2 = 1$ has no rational solution. *Hint:* Proceed by contradiction, setting $x = a/b$ for some $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^+$. Try arguing about the parity of a and b .

Relations on Sets

Given a set S , a *relation* on S is a way of comparing two elements of the set; or rather, specifying their relationship. For example, equality is a relation, for when we write $a = b$ we are specifying a relationship between a and b . Similarly, $a < b$ is a relation. Abstractly, if $a, b \in S$ and there isn't a better symbol, we write the relation as aRb or $R(a, b)$. Other common notation includes $a \sim b$, or the use of a specific symbol such as the equal or less-than sign.

Definition 7

If A and B are sets, a *binary relation* R between A and B is a subset $S_R \subseteq A \times B$. We say that A is the *domain* of the relation, and B is the *codomain*. If $a \in A$ and $b \in B$, we say that a is *related to* b (written aRb) if $(a, b) \in S_R$. When $B = A$, we simply say that R is a *relation on* A .

If $A = \{2, 4, 6\}$ and $B = \{\heartsuit, \diamond\}$, we could define a binary relation R by specifying the subset $S = \{(2, \heartsuit), (4, \heartsuit), (6, \diamond)\}$. In this case we would write $2R\heartsuit$, $4R\heartsuit$, and $6R\diamond$.

Many binary relations compare elements in the same set; that is, we take $B = A$ and so define a relation by specifying a subset of $A \times A$. For example, if $A = \{2, 4, 6\}$, then $S = \{(2, 4), (2, 6), (4, 6)\} \subseteq A \times A$ is a relation on A . If we denote this relation by R , we would say that $2R4$, $2R6$, and $4R6$. Other familiar relations we could define on A include

- Equals “=”, given by the subset $S_{=} = \{(2, 2), (4, 4), (6, 6)\}$. We would write this as $2 = 2$, $4 = 4$, and $6 = 6$.
- Less than “<”, given by the set $S_{<} = \{(2, 4), (2, 6), (4, 6)\}$ above. We would write $2 < 4$, $2 < 6$ and $4 < 6$.
- We could define the relation “ $\leq$ ” on A , in which case the set would be

$$S_{\leq} = \{(2, 2), (2, 4), (2, 6), (4, 4), (4, 6), (6, 6)\}.$$

On larger sets, it's impractical to write S down explicitly but it is nonetheless present. For example, if $A = B = \mathbb{Z}$ and R is the less-than relation " $<$ ", then $S_R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : a < b\}$. Here we see that $(-2, 5) \in S_R$ since $-2 < 5$, while $(15, 14) \notin S_R$ since $15 \not< 14$.

Exercise 8 Let A be an arbitrary non-empty set. Suppose we want to define the equality relation " $=$ " in the usual sense. What subset $S \subseteq A \times A$ gives this relation?

To distinguish between different types of relations, we define properties that the relation can exhibit.

Definition 9

If R is a relation on A and B , we say that R is

- *left-total* if for every $a \in A$ there exists $b \in B$ such that aRb .
- *functional* if for every $a \in A, b, c \in B$, if aRb and aRc then $b = c$.

When R is a relation on A (so $B = A$) then we say that R is

- *reflexive* if for all $a \in A$, aRa .
- *irreflexive* if for all $a \in A$, $a \not R a$.
- *symmetric* if for all $a, b \in A$, aRb implies that bRa .
- *anti-symmetric* if for all $a, b \in A$, aRb and bRa implies $a = b$.
- *transitive* if for all $a, b, c \in A$, aRb and bRc implies that aRc .
- *total* if for all $a, b \in A$ either $a = b$, aRb , or bRa .

For example, equality $a = b$ is reflexive ($a = a$ is always true), transitive (if $a = b$ and $b = c$ then $a = c$), and symmetric (when $a = b$ then $b = a$), but not total. On the other hand, the "less-than-or-equal" relation ($\leq$) is reflexive ($a \leq a$ is always true), transitive (if $a \leq b$ and $b \leq c$ then $a \leq c$), anti-symmetric ($a \leq b$ and $b \leq a$ implies $a = b$), and total (either $a \leq b$ or $b \leq a$).

In practice, we often define a relation in terms of how the two objects interact, just as we did above by saying that aRb if $a \leq b$.

Example 10

Let $\mathbb{R}^+$ denote the set of positive real numbers and define $a \sim b$ if $ab > 1$. Determine whether $\sim$ is reflexive, symmetric, transitive, or total.

Solution. We claim that the relation is symmetric, but not reflexive, transitive, or total. To see that the relation is symmetric, suppose that a and b are positive real numbers such that $a \sim b$; that is, $ab > 1$. But then $ab = ba > 1$, and so $b \sim a$.

To disprove the other three properties, we need only give counterexamples. For example, to show that $\sim$ is not reflexive, we take $a = 0.5$. It follows that $a^2 = 0.25 \not> 1$, and so $a \not\sim a$. The relation is not total, since if $a = 0.5$ and $b = 1$, then $ab = 0.5$ and so neither $a \sim b$ nor $b \sim a$. Finally, the relation fails to be

transitive. Setting $a = c = 0.5$ and $b = 10$, we can quickly check that $a \sim b$ and $b \sim c$. However, $a \not\sim c$, since $ac = 0.25 < 1$. ■

Exercise 11 Let R be the relationship on $A = \{2, 4, 6\}$ and $B = \{\heartsuit, \diamond\}$ mentioned above. Show that R is left-total and functional.

Exercise 12 Define a relation on $\mathbb{Z}$ by saying that aRb if there is an $n \in \mathbb{Z}$ such that $b = na$; namely, a is related to b if a divides b . Determine whether R is reflexive, (anti-)symmetric, transitive, or total.

Equivalence Relations

Definition 13

A relation R on a set A is said to be an *equivalence relation* if R is reflexive, transitive, and symmetric.

Equivalence relations are so named because equality “=” is an equivalence relation. The idea is to capture the idea of two things being morally equal, even when they’re not actually equal.

Example 14

Define a relation on $\mathbb{R}$ as $a \sim b$ if $b - a = 2\pi k$ for some $k \in \mathbb{Z}$. Show that $\sim$ is an equivalence relation.

Solution. We need to check that $\sim$ is reflexive, transitive, and symmetric.

- To show that the relation is reflexive, let $x \in \mathbb{R}$ be arbitrary. Note that $x - x = 0 = 2\pi(0)$, so that $x \sim x$.
- To show transitivity, suppose that $x, y, z \in \mathbb{R}$ are such that $x \sim y$ and $y \sim z$. Let $k_1, k_2 \in \mathbb{Z}$ be such that $y - x = 2\pi k_1$ and $z - y = 2\pi k_2$. Now

$$z - x = (z - y) + (y - x) = 2\pi k_1 + 2\pi k_2 = 2\pi(k_1 + k_2)$$

showing that $x \sim z$.

- Finally, for symmetry, suppose that $x, y \in \mathbb{R}$ are such that $x \sim y$. Let $k \in \mathbb{Z}$ be such that $y - x = 2\pi k$. But then $x - y = 2\pi(-k)$, showing that $y \sim x$ as required. ■

Exercise 15 Define a relation on $\mathbb{Z}^+ \times \mathbb{Z}^+$ as $(a, b) \sim (c, d)$ if $a + d = b + c$. Show that $\sim$ is an equivalence relation.

Definition 16

Given an equivalence relation $\cong$ on a set A , the *equivalence class* of an element $a \in A$ is the set

$$[a] = \{x \in A : x \cong a\}.$$

If $\cong$ is an equivalence relation on S , the set of equivalence classes in S is denoted by the quotient $S/\cong$; that is, $S/\cong = \{[a] : a \in A\}$. There is a complicated reason for this that is beyond the scope of this course.

Example 17

Let $\sim$ be the equivalence relation from Example 14. What do elements of an equivalence class look like?

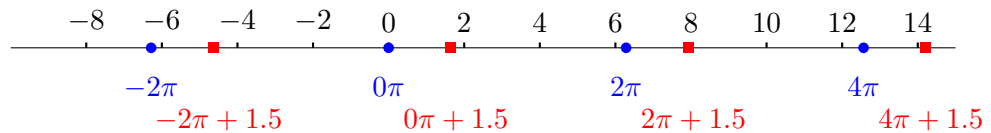
Solution. Let's start with a simple example and look at the equivalence class $[0]$. By definition,

$$[0] = \{x \in \mathbb{R} : x \sim 0\} = \{x \in \mathbb{R} : x = 2\pi k, k \in \mathbb{Z}\},$$

so the equivalence class of $[0]$ consists of the integer multiples of 2π . What about something like $[1.5]$? Here we have

$$[1.5] = \{x \in \mathbb{R} : x \sim 1.5\} = \{x \in \mathbb{R} : x - 1.5 = 2\pi k, k \in \mathbb{Z}\}.$$

Thus if $y \in [1.5]$, then $y = 1.5 + 2\pi k$ for some $k \in \mathbb{Z}$; that is, they look like the elements of $[0]$ only shifted 1.5 units to the right. The blue circles below denote $[0]$ while the red squares denote $[1.5]$.



In general, the equivalence classes $[x]$ are all those real numbers that are multiples of 2π away from one another. ■

Exercise 18 Let $\cong$ be an equivalence relation on A . Show that

1. For any $a \in A$, $a \in [a]$.
2. If $a, b \in A$, then $[a] = [b]$ if and only if $a \cong b$.
3. If $a, b \in A$ then $[a] \cap [b] = \emptyset$ if and only if $a \not\cong b$.

Theorem 19

If $\sim$ is an equivalence relation on A , then $\sim$ partitions S into disjoint equivalence classes; that is, every element $x \in S$ belongs to a unique equivalence class.

Solution. You've already proven this via Exercise 18! In part (i) you showed that every element of A belongs to some equivalence class. In part (ii) you showed that related elements define the same equivalence class, and hence equivalence classes are unique. And in part (iii) you showed that disjoint equivalence classes have no overlap. Thus equivalence classes perfectly partition A . ■

Order Relations

Having generalized the notion of equality, we next turn our attention to generalizing the notion of inequality. As mentioned previously, we know that " $\leq$ " is reflexive, transitive, and anti-symmetric. However, if we wanted to generalize the notion of a strict inequality " $<$ ", then we would be irreflexive instead of reflexive. Thus we have two definitions, as follows:

Definition 20

A relation R on a set A is said to be a *weak (strong) order relation*^a if R is transitive, anti-symmetric, and reflexive (irreflexive). We say that R is a *linear or total order on A* if it is total, and a *partial order* otherwise.

^aTry to think about why the words strong and weak are used here.

The set A together with a partial order relation R is called a *partially ordered set*, or a *poset* for short. We often write posets as pairs (A, R) . While the majority of order relations with which you are already familiar are total, it is fairly easy to come up with natural examples of partial orders.

Example 21

Let X be a set and define a relation on the power set $\mathcal{P}(X)$ by subset inclusion; namely, ARB if $A \subseteq B$. Show that R is a weak partial order relation.

Solution. This relation is reflexive since $A \subseteq A$ is always true. It is anti-symmetric, for if $A \subseteq B$ and $B \subseteq A$ then $A = B$, by definition. It is also transitive, since if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

However, notice that inclusion is *not* total. Given two arbitrary subsets $A, B \in \mathcal{P}(X)$, there need not be a relation between them. It could be the case that $A \subseteq B$ or $B \subseteq A$, but in general neither need be true. For example, $A = \{1, 2\}$ and $B = \{2, 3\}$ satisfies neither $A \subseteq B$ nor $B \subseteq A$. ■

Definition 22

Given a weak partially ordered set $(A, \preceq)$, we say that $a \in A$ is a *maximal* element of A when for any $x \in A$, if $a \preceq x$ then $a = x$. Similarly, a is a *minimal* element of A if $x \preceq a$ implies that $x = a$.

Maximal and minimal elements capture the idea of being the “biggest” and “smallest” elements, respectively. Note however that these are different from being the *maximum* and *minimum*, as we’ll discuss shortly.

Example 23

Consider the set $A = \{1, 2, 3, 5, 6, 10, 15\}$ with the relation $a \preceq b$ if there is some $n \in \mathbb{Z}$ such that $b = na$. Find the maximal elements in A .

Solution. You have already shown in Exercise 12 that this is a weak partial order. The maximal elements are the “biggest” elements, in the sense that there is no larger element of A which our number divides. If you think about this for a second, you will see that the natural candidates for this are 6, 10, and 15 (why?).

Let’s prove then that 6 is maximal. Indeed, suppose that $x \in A$ satisfies $6 \preceq x$; that is, there is an $n \in \mathbb{Z}$ such that $x = 6n$. Consequently, x must be a multiple of 6, and the only element in A which is a multiple of 6 is $x = 6$. Thus $x = 6$, as required. The proofs for 10 and 15 are similar. ■

Exercise 24 Finish the proof from Example 23 that 10 and 15 are maximal.

Exercise 25 What are the minimal elements from Example 23?

The *maximum* of a weak partially ordered set $(A, \preceq)$ is an element $M \in A$ such that $x \preceq M$ for all $x \in A$. Note that this is different from a maximal element, since the maximum must be larger than *every* element in the set; namely, it must be comparable to every element in the set. Similarly, the *minimum* is an element $m \in A$ such that $m \preceq x$ for all $x \in A$ – it is smaller than every element of the set. The number 1 is the minimum of Example 23; however, there is no maximum.

Exercise 26 Show that there is no maximum element in Example 23. What is the smallest number you could add to A that would be a maximum?

Definition 27

Given a weak partially ordered set $(A, \preceq)$ and a subset $S \subseteq A$, we say that $M \in A$ is an *upper bound* of S if $x \preceq M$ for all $x \in S$. We say that $m \in A$ is a *lower bound* of S if $m \preceq x$ for all $x \in S$.

For example, in Example 23, the set $S = \{2, 3\}$ has an upper bound of 6 and a lower bound of 1. However, the set $\{2, 3, 5\}$ contains no upper bound! Example 23 doesn’t have enough numbers to really make our case though, so let’s expand our set.

Example 28

Consider the set $\mathbb{Z}^+$ with the relation $a \preceq b$ if there exists some $n \in \mathbb{Z}^+$ such that $b = na$. Give three lower bounds and three upper bounds for the set $S = \{8, 12, 36\}$.

Solution. Let's start with the lower bounds. We're looking for numbers which divide all three of the elements of S . Clearly 1 works, as do 2 and 4. In fact, this completely exhausts the list of lower bounds, since the only other number which divides 8 is 8 itself, and that does not divide 12 or 36.

On the other hand, there are infinitely many upper bounds. We want to find three numbers, each of which contains 8, 12, and 36 as divisors. The number 72 is the smallest such upper bound, and once we have an upper bound, we can just take any multiple of that upper bound to create more upper bounds. Thus 144 and 216 are both also upper bounds. ■

Exercise 29 In the previous example, we made two claims: That 72 is the smallest upper bound for S , and that if n is an upper bound for S , then so too is nk for any $k \in \mathbb{Z}^+$. Prove these facts.

We finish with the following definitions, which are extremely important in calculus and will be useful to us down the road.

Definition 30

Given a weak partially ordered set $(A, \preceq)$ and a set $S \subseteq A$, the *supremum* of S – denoted $\sup(S)$ – is the least upper bound of S ; namely, it is an upper bound M for S such that $M \preceq U$ for all other upper bounds U of S . Similarly, the *infimum* of S – denote $\inf(S)$ – is the greatest lower bound; that is, a lower bound m such that $\ell \preceq m$ for all lower bounds ℓ of S .

As we mentioned in Example 28, the infimum of S in that problem was $\inf(S) = 4$, while the supremum was $\sup(S) = 72$.

Exercises

- Prove the following by contradiction:
 - If x and y are real numbers whose sum $x + y$ is irrational, then at least one of x or y must be irrational.
 - If p is a prime number greater than 2, then p is odd.
 - If p is a prime number, then $\sqrt{p}$ is irrational.
- For each relation below, determine which of the properties from Definition 9 are satisfied.
 - The relation on $\mathbb{Z}$ defined by $a \sim b$ if $|a - b|$ is even.
 - The relation on $\mathbb{Q}$ defined by $a \sim b$ if $ab \in \mathbb{Z}$.

- (c) The relation on $\mathbb{Z}^+$ defined by $a \sim b$ if $1/a < 1/b$.
 - (d) The relation on $\mathbb{Z}$ defined by $a \sim b$ if the number of digits in a is less than the number of digits in b . For example, $12 \sim 523$.
 - (e) The relation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \sim (c, d)$ if $a + b \leq c + d$.
 - (f) The relation on $\mathbb{R}$ defined by $a \sim b$ if $a - b \in \mathbb{Q}$
 - (g) The relation on $\mathbb{R}$ defined by $a \sim b$ if $|a| = |b|$.
3. Which of the relations from Exercise 2 is an equivalence relation?
 4. Which of the relations from Exercise 2 is an order relation?
 5. The sgn-function assigns to each real number its corresponding sign; namely,

$$\text{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}.$$

Define a relation on $\mathbb{R}$ by $a \sim b$ if $\text{sgn}(a) = \text{sgn}(b)$.

- (a) Show that $\sim$ is an equivalence relation.
 - (b) Into how many equivalence classes does $\sim$ partition $\mathbb{R}$?
6. Define a relation on $\mathbb{Z} \times \mathbb{Z}^+$ by $(a, b) \sim (c, d)$ if $ad = bc$.
 - (a) Show that $\sim$ is an equivalence relation.
 - (b) Give 3 elements of the equivalence class $[(1, 2)]$.
 - (c) We can define the sum of equivalence classes as $[(a, b)] + [(c, d)] = [(ad + bc, bd)]$. Show that this is well-defined; namely, that this does not depend on the representative chosen from each equivalence class.
 - (d) We can define the product of equivalence classes as $[(a, b)] \times [(c, d)] = [(ac, bd)]$. Show that this is well-defined; namely, that this does not depend on the representative chosen from each equivalence class.
 - (e) What have we just defined?
 7. A set $S \subseteq \mathbb{Z}^+$ is said to be *divisor-closed* if for every element $x \in S$, every divisor of x is also in S . For example, the set $S = \{1, 2, 3, 4, 6, 12\}$ is divisor closed. Let $U = \{1, 2, 3, 4, 5, 6, 12\}$.
 - (a) Let D denote the set of divisor-closed subsets of U . Write out all the elements of D .
 - (b) Define a partial order on D by inclusion; namely, $A \preceq B$ if $A \subseteq B$.
 - (c) What are the maximal and minimal elements of D ?
 - (d) Let $S = \{\{1, 2\}, \{1, 3\}\}$. Find all the upper bounds for S . What is $\sup(S)$?
 - (e) Let S be as defined in part 7d. Find all the lower bounds for S . What is $\inf(S)$?